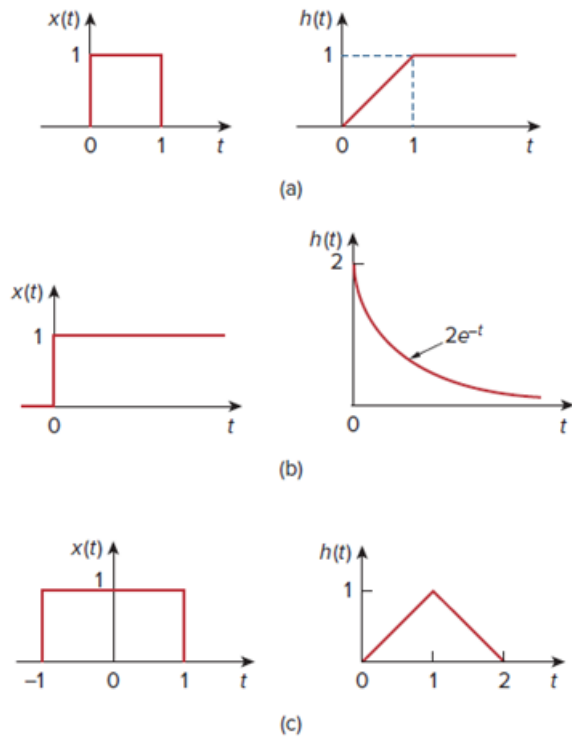


Problem

Find $y(t) = x(t) * h(t)$ for each paired $x(t)$ and $h(t)$ in Fig. 15.37.

Figure 15.37:

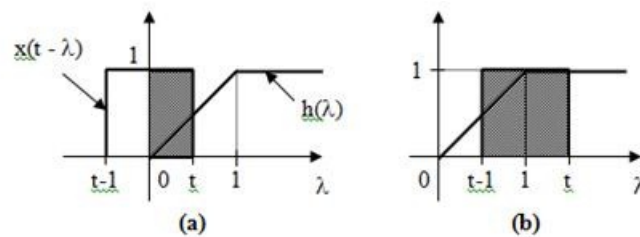


Step-by-step solution

Step 1 of 1

For $0 < t < 1$, $x(t - \lambda)$ and $h(\lambda)$ overlap as shown in Fig. (a).

$$y(t) = x(t) * h(t) = \int_0^t (1)(\lambda) d\lambda = \frac{\lambda^2}{2} \Big|_0^t = \frac{t^2}{2}$$

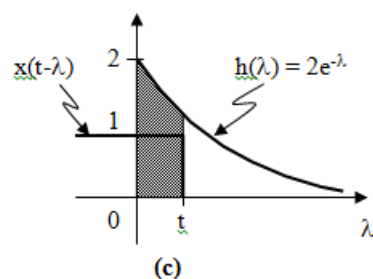


For $1 < t < 2$, $x(t - \lambda)$ and $h(\lambda)$ overlap as shown in Fig. (b).

$$y(t) = \int_{t-1}^1 (1)(\lambda) d\lambda + \int_1^t (1)(1) d\lambda = \frac{\lambda^2}{2} \Big|_{t-1}^1 + \lambda \Big|_1^t = \frac{-1}{2} t^2 + 2t - 1$$

For $t > 2$, there is a complete overlap so that

$$y(t) = \int_{t-1}^t (1)(1) d\lambda = \lambda \Big|_{t-1}^t = t - (t-1) = 1$$



Therefore,

$$y(t) = \begin{cases} t^2/2, & 0 < t < 1 \\ -(t^2/2) + 2t - 1, & 1 < t < 2 \\ 1, & t > 2 \\ 0, & \text{otherwise} \end{cases}$$

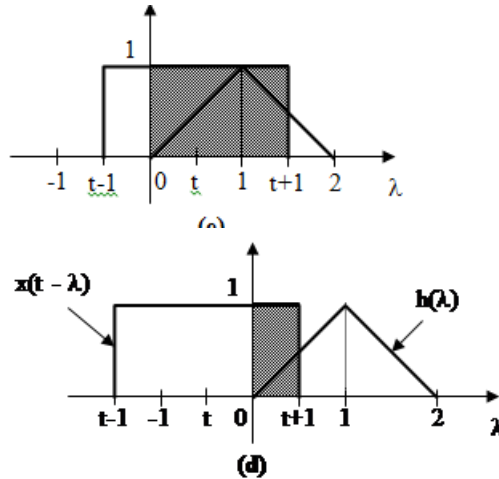
(b) For $t > 0$, the two functions overlap as shown in Fig. (c).

$$y(t) = x(t) * h(t) = \int_0^t (1) 2e^{-\lambda} d\lambda = -2e^{-\lambda} \Big|_0^t$$

Therefore,

$$y(t) = 2(1 - e^{-t}), \quad t > 0$$

(c) For $-1 < t < 0$, $x(t - \lambda)$ and $h(\lambda)$ overlap as shown in Fig. (d).

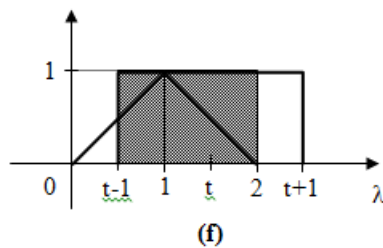


$$y(t) = x(t) * h(t) = \int_0^{t+1} (1)(\lambda) d\lambda = \frac{\lambda^2}{2} \Big|_0^{t+1} = \frac{1}{2}(t+1)^2$$

For $0 < t < 1$, $x(t - \lambda)$ and $h(\lambda)$ overlap as shown in Fig. (e).

$$y(t) = \int_0^1 (1)(\lambda) d\lambda + \int_1^{t+1} (1)(2 - \lambda) d\lambda$$

$$y(t) = \frac{\lambda^2}{2} \Big|_0^1 + \left(2\lambda - \frac{\lambda^2}{2} \right) \Big|_1^{t+1} = \frac{1}{2}t^2 + t + \frac{1}{2}$$

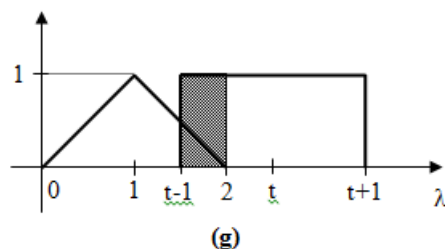


For $1 < t < 2$, $x(t - \lambda)$ and $h(\lambda)$ overlap as shown in Fig. (f).

$$y(t) = \int_{t-1}^1 (1)(\lambda) d\lambda + \int_1^2 (1)(2 - \lambda) d\lambda$$

$$y(t) = \frac{\lambda^2}{2} \Big|_{t-1}^1 + \left(2\lambda - \frac{\lambda^2}{2} \right) \Big|_1^2 = \frac{1}{2}t^2 + t + \frac{1}{2}$$

For $2 < t < 3$, $x(t - \lambda)$ and $h(\lambda)$ overlap as shown in Fig. (g).



$$y(t) = \int_{t-1}^2 (1)(2 - \lambda) d\lambda = \left(2\lambda - \frac{\lambda^2}{2} \right) \Big|_{t-1}^2 = \frac{9}{2} - 3t + \frac{1}{2}t^2$$

Therefore,

$$\int (t^2/2) + t + 1/2, \quad -1 < t < 0$$

$$y(t) = \begin{cases} (t^2/2) + t + 1/2, & 0 < t < 2 \\ (t^2/2) - 3t + 9/2, & 2 < t < 3 \\ 0, & \text{otherwise} \end{cases}$$